Traffic Sign Recognition System

System Architecture Specification & Technical Documentation — Category: Computer Vision & Autonomous Systems

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1. Autonomous Driving & Road Sign Classification

Automated recognition of traffic signs is essential for Advanced Driver Assistance Systems (ADAS) and autonomous vehicles. The system specifies deep convolutional neural network classification across standard traffic sign categories (Speed limits, Stop, Yield, Danger warnings, Prohibitions) using the German Traffic Sign Recognition Benchmark (GTSRB).

2. Proposed Computer Vision Pipeline

• **Image Preprocessing:** Histogram equalization, adaptive contrast enhancement (CLAHE), and image resizing to canonical dimensions (32x32).

• **Data Augmentation:** Random rotations, zooming, and shearing to simulate varying weather and lighting conditions.

• **CNN Architecture:** Multi-layer Conv2D + MaxPooling with Dropout regularization and Softmax multi-class output.

• **Inference:** Streamlit dashboard supporting live image upload and video stream analysis.

3. Verification Status

• **Status:** Needs verification. Clearly labeled as pending in portfolio.

Architecture Note: Repository link is maintained as pending per instructions to ensure full verification integrity.